

Camouflage-Aware Medical Image Segmentation

1. Objective

Build a deep learning model that segments medical images by treating lesions as camouflaged objects. The goal is to improve detection of low-contrast or hidden regions.

The impact of camouflaged trained model in segmentation of medical images.

2. Basic Idea

Lesions often blend with surrounding tissue. We add a camouflage detection branch and a simple enhancement (counterfactual) mechanism to help the segmentation model focus on hidden regions.

3. Architecture

- Use U-Net as base model
- Add Camouflage Branch: predicts difficult regions
- Enhance features: combine original + camouflage features ()
- Final segmentation output

4. Steps to Implement

Step 1: Load dataset (e.g., Kvasir-SEG, COD10K)

Step 2: Preprocess images (resize, normalize)

Step 3: Build U-Net model / Attention U-Net

Step 4: Add camouflage branch

Step 5: Fuse features

Step 6: Train using Dice + BCE loss

Step 7: Evaluate using Dice score and IoU

Dataset Link:

Training :

1. COD10K: <https://www.kaggle.com/datasets/ivanomelchenkoim11/cod10k-dataset>
2. Kvasir-SEG: <https://www.kaggle.com/datasets/debeshjha1/kvasirseg>

Testing:

3. <https://challenge.isic-archive.com/data/#2016>

1. Project Overview

The objective is to evaluate the impact of **Camouflage Object Detection (COD)** features on **Medical Image Segmentation**. The core hypothesis is that a model trained to identify hidden animals or objects in nature (camouflage) will provide superior feature maps for identifying low-contrast or blended lesions in medical scans.

2. Datasets Used

The research utilizes three distinct datasets to ensure a clean separation between camouflage learning and medical application:

- **CD1 (Camouflage Dataset 1):** Typically **COD10K**, used to teach the model how to identify objects that blend into their background.
- **MD1 (Medical Dataset 1):** Typically **Kvasir-SEG**, used for the primary guided training of the medical model.
- **MD2 (Medical Dataset 2):** Typically **ISIC (Skin Lesion)** or similar, used as a completely unseen testing set to measure the final performance impact.

Results Comparison :

Model Code	Model Description	Training Dataset	Pre-training / Guidance	Testing Dataset	Key Role in Research
M1	Standard	Medical	None	Medical	Baseline: Shows

	Medical U-Net	(MD1)		(MD2)	performance without any special features.
MUA1	Medical Attention U-Net	Medical (MD1)	Internal Attention Gates	Medical (MD2)	Industry Benchmark: Shows standard way to improve focus.
CU1	Camouflage U-Net	Camouflage (CD1)	None	Medical (MD2)	Cross-Domain Test: Can a "nature" model find "lesions" directly?
CUA1	Camouflage Attention U-Net	Camouflage (CD1)	Internal Attention Gates	Medical (MD2)	Cross-Domain Benchmark: Does attention help a nature model on medical data?
G-UNet 1	Guided U-Net	Medical (MD1)	Guided by CU1 Features	Medical (MD2)	Your Innovation: Shows impact of camouflage features on medical training.
G-UNet 2	Guided U-Net	Medical (MD1)	Guided by CUA1 Features	Medical (MD2)	Your Innovation: Shows impact of camouflage features on medical training.
G-AUNet 1	Guided	Medical	Guided by	Medical	The Final Model:

	Attention U-Net	(MD1)	CU1 Features	(MD2)	Combines camouflage knowledge with spatial filtering.
G-AUNet	Guided Attention U-Net	Medical (MD1)	Guided by CUA1 Features	Medical (MD2)	The Final Model: Combines camouflage knowledge with spatial filtering.

3. Three-Stage Methodology

Stage 1: Pre-training the Camouflage Experts (CU1 & CUA1)

- **Action:** Construct two baseline architectures: a standard U-Net and an Attention U-Net.
- **Training:** Both models are trained exclusively on the **Camouflage Dataset (CD1)**.
- **Goal:** To develop "expert" feature extractors capable of identifying high-level "hidden" patterns, such as subtle textures and low-contrast boundaries in nature.
- **Result:** Upon completion, the weights are saved as **CU1** (Standard Camouflage) and **CUA1** (Attention-Refined Camouflage).

Stage 2: Guided Training of the Medical Models (G1–G4)

- **Action:** Initialize the training of the Medical U-Nets on **MD1** (Medical Dataset 1).
- **Guidance Mechanism:** The pre-trained camouflage models (**CU1** or **CUA1**) are integrated into the pipeline in a **"Frozen" state**, meaning their weights remain static while they act as feature consultants.

- **Layer-by-Layer Feature Guidance:** At each encoder level, feature maps from the frozen camouflage expert are extracted and fused with the active medical encoder.
 - **Logic:** $\text{Encoder}_{\text{Medical}}(\text{Level } n) + \text{Feature}_{\text{Camouflage}}(\text{Level } n) \rightarrow \text{Decoder}_{\text{Medical}}(\text{Level } n)$
- **Ablation Variants:**
 - **G1 & G2:** Guide a standard Medical U-Net using **CU1** and **CUA1** respectively.
 - **G3 & G4:** Guide a Medical Attention U-Net using **CU1** and **CUA1** respectively.
- **Loss Function:** Models are optimized using a combined **Dice Loss + Binary Cross Entropy (BCE)** to ensure high-precision segmentation of lesion boundaries.

Stage 3: Testing and Comparative Analysis

- **Testing:** All trained models (**M1**, **MUA1**, **G1**, **G2**, **G3**, **G4**) and the direct cross-domain models (**CU1**, **CUA1**) are evaluated on the unseen **Medical Dataset (MD2)**.
- **Comparative Baseline:** Results are benchmarked against **M1** (Standard Medical U-Net) and **MUA1** (Industry Standard Attention U-Net) to quantify the specific performance gain provided by camouflage knowledge.
- **Metrics:** Performance is measured using **Dice Similarity Coefficient (DSC)** and **Intersection over Union (IoU)** to assess spatial accuracy and overlap.

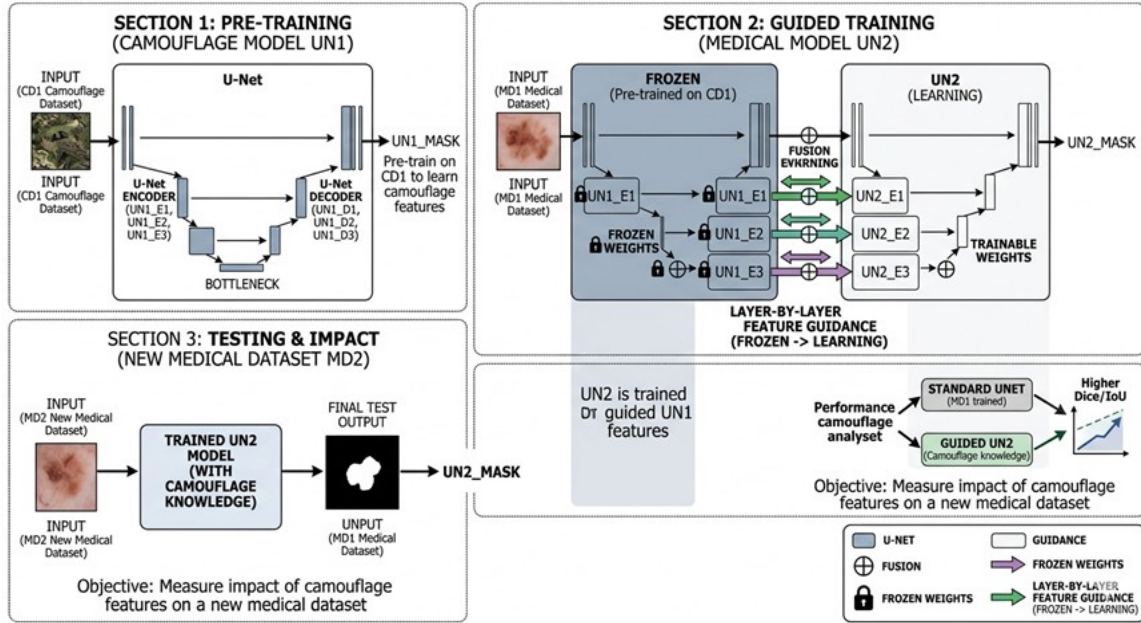


Figure 1 : G-UNet Model guided by the features of CU1

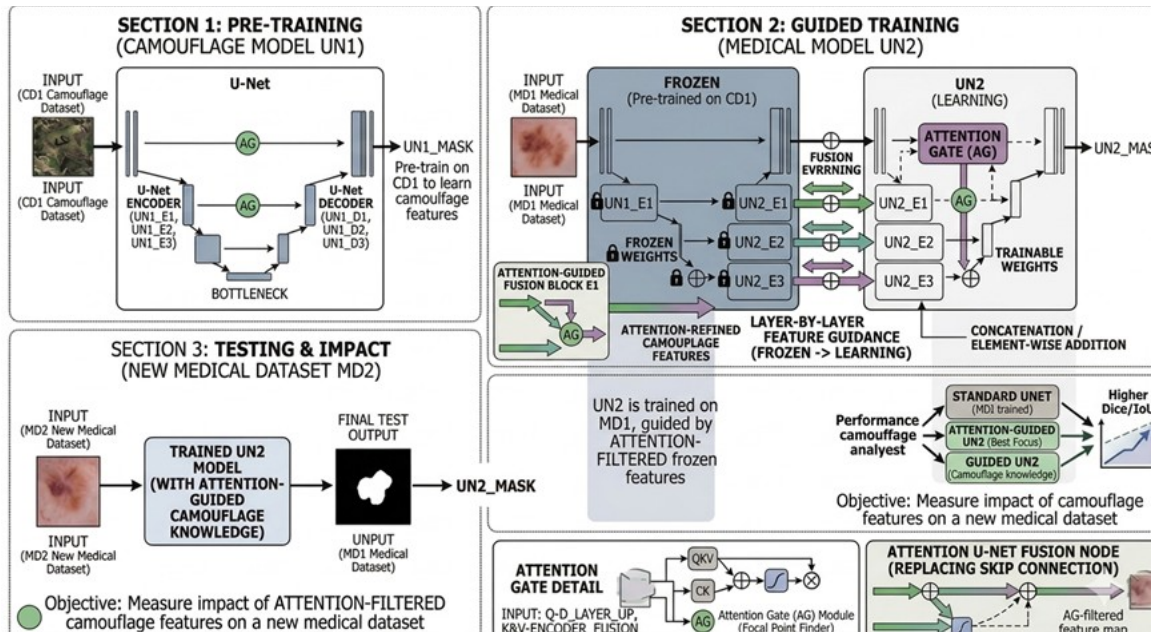


Figure 2 : G-AUNet Model guided by the features of CUA1.

5. Simple Loss Function

Total Loss = Dice Loss + Binary Cross Entropy Loss

6. Tools Required

- Python
- PyTorch / TensorFlow
- OpenCV
- NumPy

7. Expected Outcome

Improved segmentation for low-contrast lesions compared to standard U-Net.